

I. Recognisability by Semigroups and Monoids

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Semigroups

$$(S, \cdot) \quad \begin{array}{l} a, b \in S \Rightarrow a \cdot b \in S \\ (a \cdot b) \cdot c = a \cdot (b \cdot c) \end{array} \quad abc \stackrel{\text{def}}{=} a \cdot b \cdot c$$

Example 1 $(\{0, 1, \dots, n-1\}, +_n)$ $x +_n y \stackrel{\text{def}}{=} x+y \bmod n$

Example 2 $(2^{A \times A}, \circ)$ $P \circ Q \stackrel{\text{def}}{=} \{(x, y) \mid \exists z \in A . (x, z) \in P \text{ and } (z, y) \in Q\}$

Example 3 $(\Sigma^+, \cdot)$ $u \cdot v \stackrel{\text{def}}{=} uv$

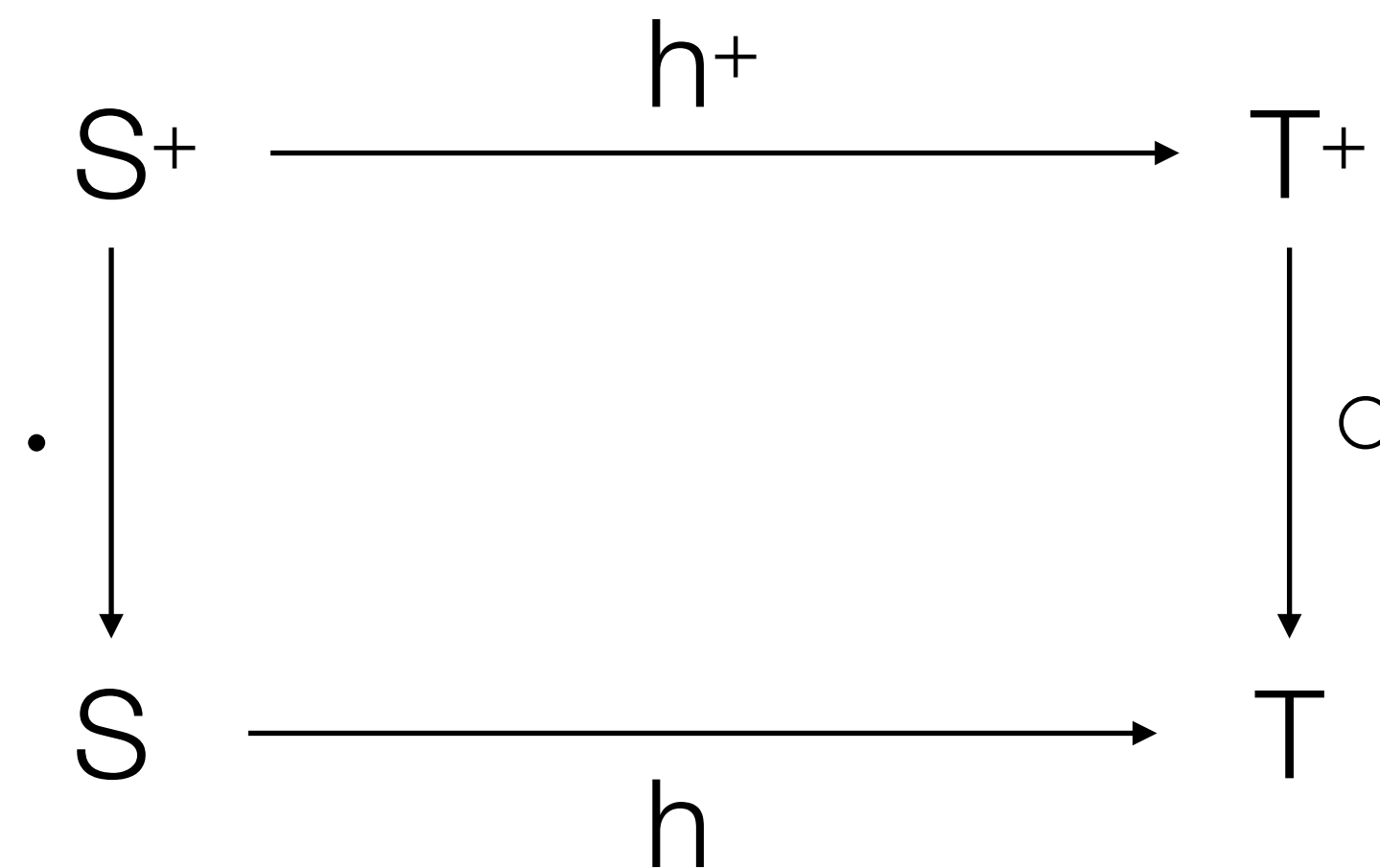
Semigroup Homomorphism

$$h : (S, \cdot) \rightarrow (T, \circ)$$

$$h(a \cdot b) = h(a) \circ h(b)$$

$S^+ \stackrel{\text{def}}{=} \text{set of non-empty sequences of elements from } S$

$$h^+(a_1, a_2, \dots, a_n) \stackrel{\text{def}}{=} h(a_1), h(a_2), \dots, h(a_n)$$



$$h(a_1 \cdot a_2 \cdot \dots \cdot a_n) = h(a_1) \circ h(a_2) \circ \dots \circ h(a_n)$$

Monoids

$$a, b \in S \Rightarrow a \cdot b \in S$$

$$(S, \textcolor{red}{1}, \cdot)$$

$$(a \cdot b) \cdot c = a \cdot (b \cdot c)$$

$$abc \stackrel{\text{def}}{=} a \cdot b \cdot c$$

$$\textcolor{red}{1} \cdot a = a \cdot \textcolor{red}{1} = a$$

Example 1

$$(\{0, 1, \dots, n-1\}, 0, +_n)$$

$$x +_n y \stackrel{\text{def}}{=} x+y \bmod n$$

Example 2

$$(2^{A \times A}, \text{Id}_A, \circ)$$

$$P \circ Q \stackrel{\text{def}}{=} \{(x, y) \mid \exists z \in A . (x, z) \in P \text{ and } (z, y) \in Q\}$$

$$\text{Id}_A \stackrel{\text{def}}{=} \{(x, x) \mid x \in A\}$$

Example 3

$$(\Sigma^*, \varepsilon, \cdot)$$

$$u \cdot v \stackrel{\text{def}}{=} uv$$

Monoid Homomorphism

$$h : (S, 1_S, \cdot) \rightarrow (T, 1_T, \circ)$$

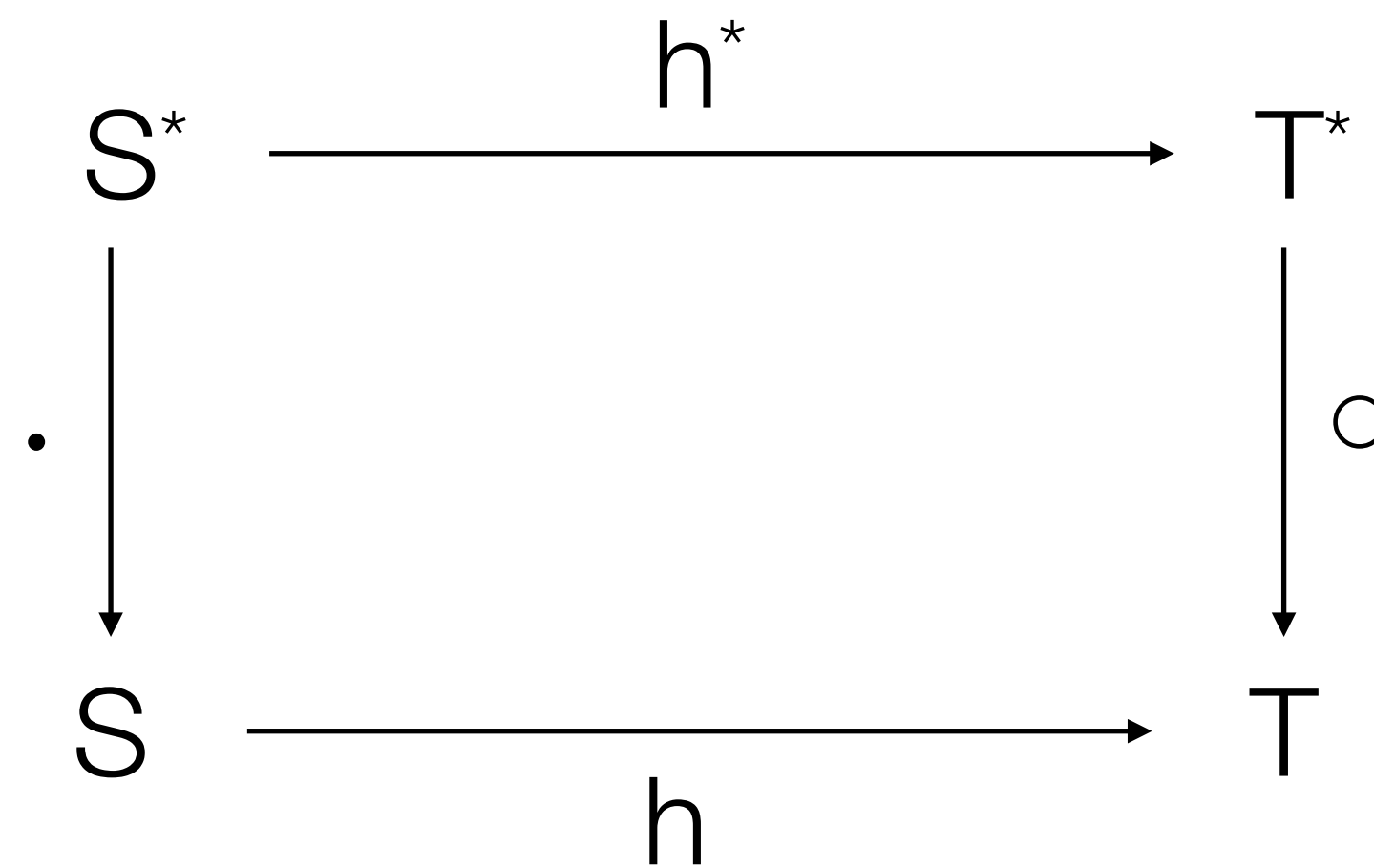
$$h(a \cdot b) = h(a) \circ h(b)$$

$S^* \stackrel{\text{def}}{=} \text{set of sequences of elements from } S$

$$h(1_S) = 1_T$$

$$h^*(a_1, a_2, \dots, a_n) \stackrel{\text{def}}{=} h(a_1), h(a_2), \dots, h(a_n)$$

$$h^*(\epsilon) \stackrel{\text{def}}{=} \epsilon$$



$$h(a_1 \cdot a_2 \cdot \dots \cdot a_n) = h(a_1) \circ h(a_2) \circ \dots \circ h(a_n)$$

Recognisability

$(S, \cdot)$ semigroup (same definition for S monoid)

S

$I \cup$

L

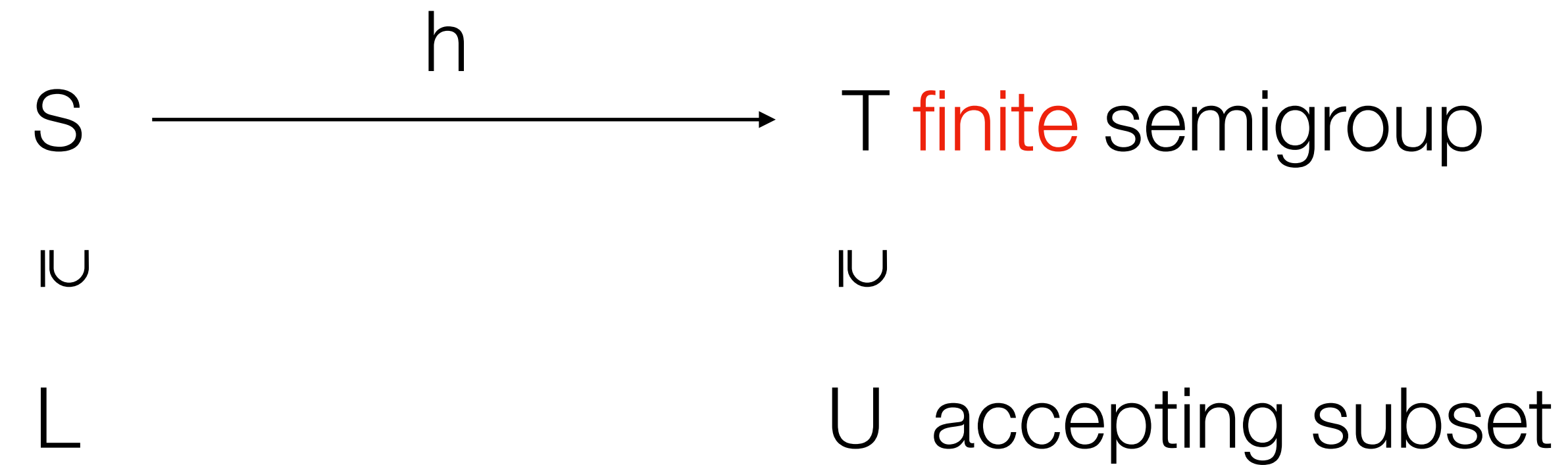
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$(S, \cdot)$ semigroup (same definition for S monoid)

$$\begin{array}{ccc} S & \xrightarrow{h} & T \text{ finite semigroup} \\ \cup & & \\ L & & \end{array}$$

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$$\begin{array}{ccc} S & \xrightarrow{h} & T \text{ finite semigroup} \\ \downarrow \text{IU} & & \downarrow \text{IU} \\ L & = & h^{-1}(U) \text{ accepting subset} \end{array}$$

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Example 1 The set of odd (even) numbers is recognisable in $(\mathbb{N}, +)$

Recognisable Word Languages

Theorem $L \subseteq \Sigma^*$ is recognisable in the monoid $(\Sigma^*, \varepsilon, \cdot)$

$\Leftrightarrow$

L is the language of a nondeterministic finite automaton $A = (Q, I, F, \rightarrow)$

" $\Rightarrow$ "

$$\begin{array}{ccc} \Sigma^* & \xrightarrow{h} & (Q, 1_Q, \circ) \text{ finite monoid} \\ \text{IU} & & \text{IU} \\ L & = & h^{-1}(F) \end{array}$$

$$A = (Q, \{1_Q\}, F, \rightarrow) \qquad q \xrightarrow{a} q \circ h(a)$$

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" $\Leftarrow$ "

$$\Sigma^* \xrightarrow{h(w)=\{(q,q') \mid q \xrightarrow{w} q'\}} (2^{Q \times Q}, \text{Id}_Q, \circ)$$

$I \cup$

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$$L = h^{-1}(\{R \in 2^{Q \times Q} \mid \exists i \in I \exists f \in F . (i, f) \in R\})$$

Syntactic Monoids

$(S, 1_S, \cdot)$ monoid $\sim \subseteq S \times S$ $a \sim b \Rightarrow sat \sim sbt$, for all $s, t \in S$ congruence

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Theorem [Myhill-Nerode]

$L \subseteq S$ is recognisable $\Leftrightarrow S/\sim_L$ is finite

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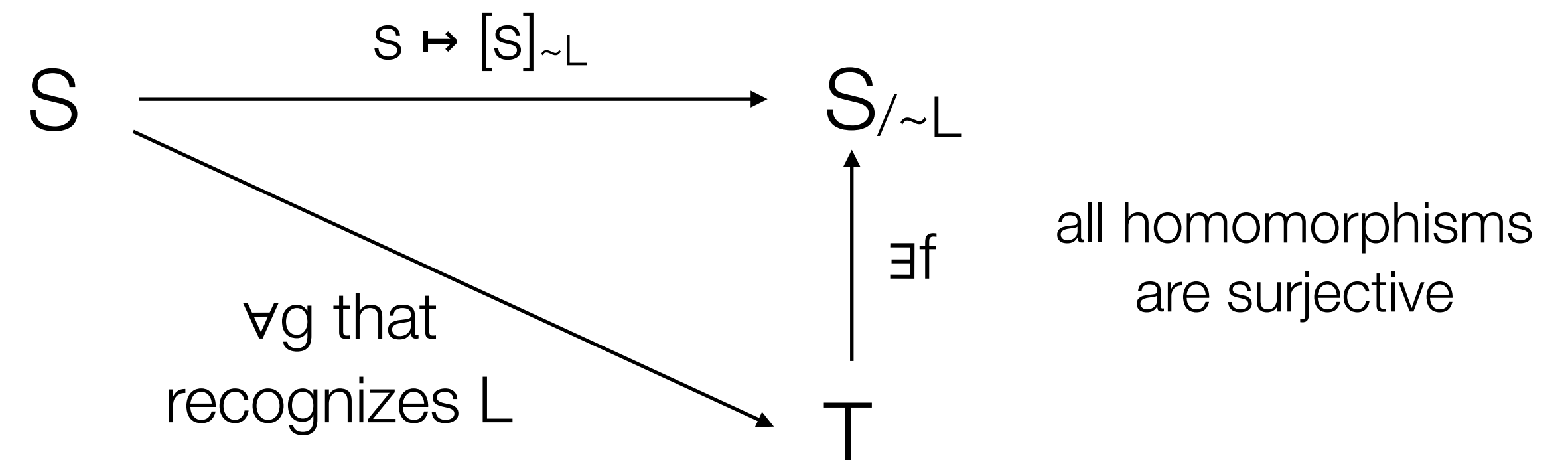
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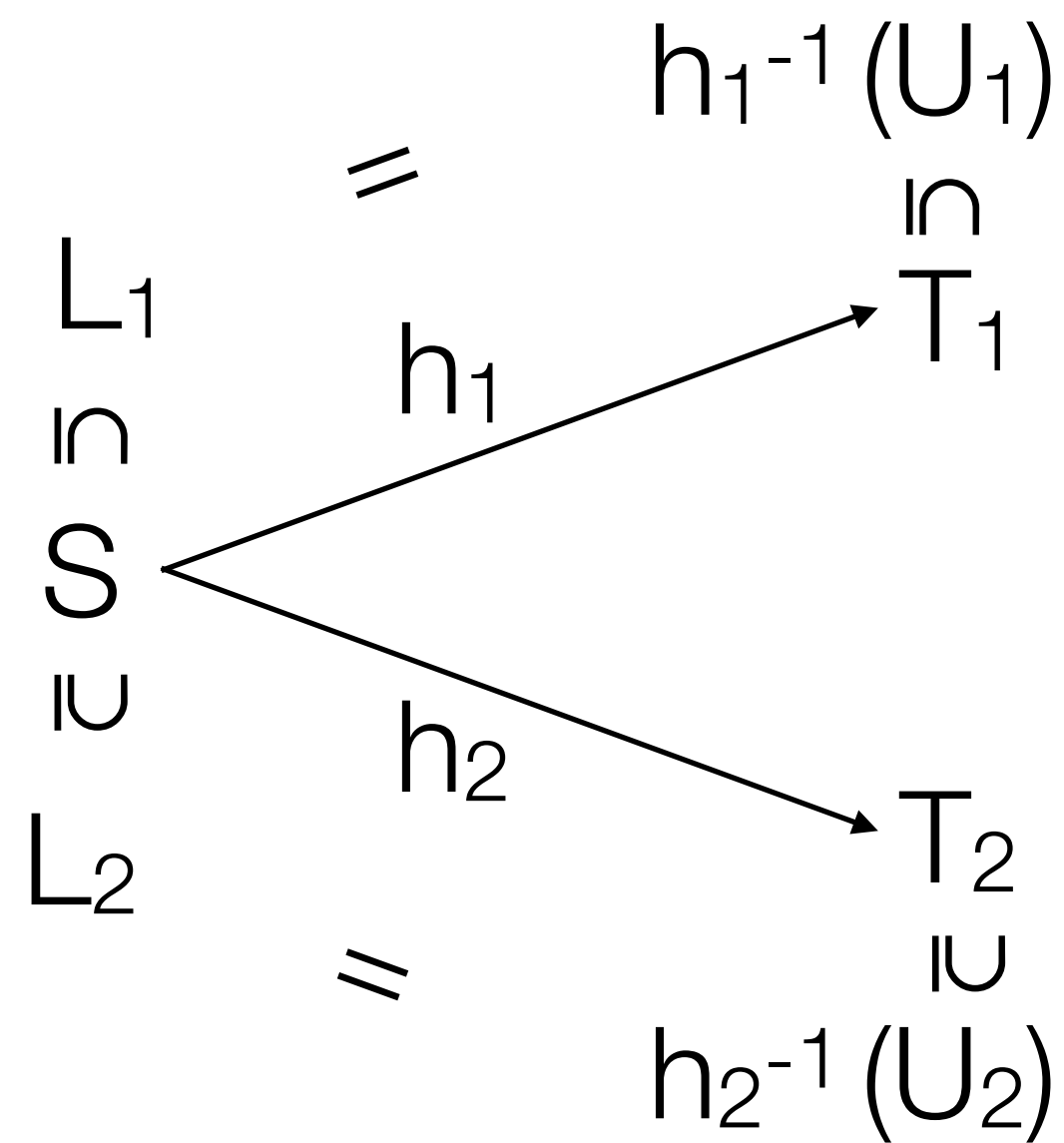
Properties of Recognisable Sets

Recognisable sets are closed under boolean combinations

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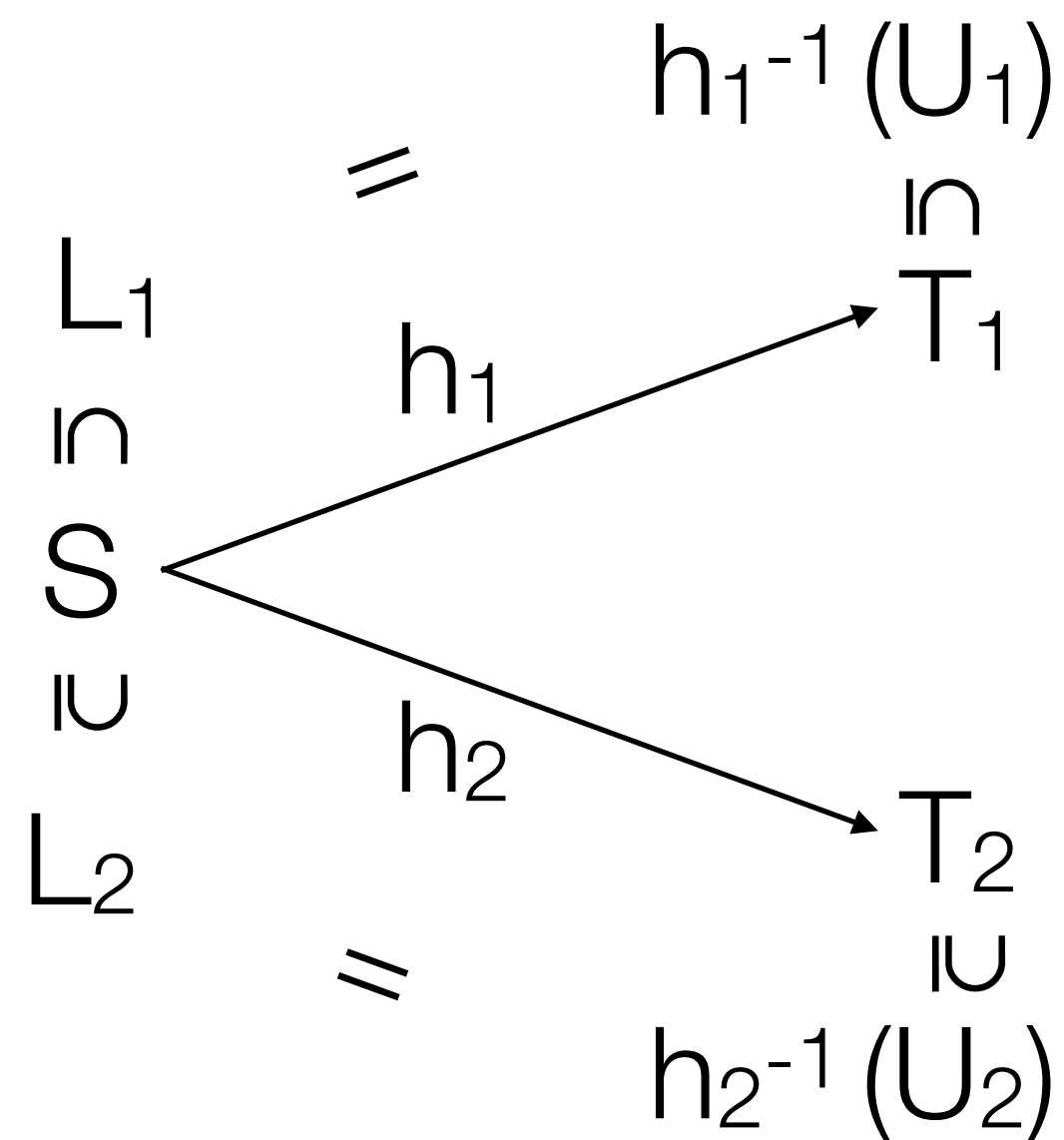
intersection



Properties of Recognisable Sets

Recognisable sets are closed under boolean combinations

intersection



$$(s_1, s_2) \otimes (t_1, t_2) \stackrel{\text{def}}{=} (s_1 t_1, s_2 t_2)$$

$$S \xrightarrow{h \stackrel{\text{def}}{=} (h_1, h_2)} (T_1 \times T_2, (1_{T_1}, 1_{T_2}), \otimes)$$

$$\begin{array}{ccc}
 \text{1U} & & \text{1U} \\
 L_1 \cap L_2 & = & h^{-1}(U_1 \times U_2)
 \end{array}$$

Properties of Recognisable Sets

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complement

$$\begin{array}{ccc} S & \xrightarrow{h} & T \\ \subseteq & & \subseteq \\ L & = & h^{-1}(U) \end{array}$$

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Recognisable sets are closed under boolean combinations

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$$\begin{array}{ccc}
 S & \xrightarrow{h} & T \\
 \sqcup & & \sqcup \\
 L & = & h^{-1}(U)
 \end{array}$$

$$\begin{array}{ccc}
 S & \xrightarrow{h} & T \\
 \sqcup & & \sqcup \\
 S \setminus L & = & h^{-1}(T \setminus U)
 \end{array}$$

Properties of Recognisable Sets

Recognisable sets are closed under inverse homomorphisms

$$\begin{array}{ccc} S & \xrightarrow{h} & T \\ \downarrow \text{IU} & & \downarrow \text{IU} \\ L & = & h^{-1}(U) \end{array}$$

Properties of Recognisable Sets

Recognisable sets are closed under inverse homomorphisms

$$\begin{array}{ccccc} S' & \xrightarrow{g} & S & \xrightarrow{h} & T \\ \downarrow \text{IU} & & \downarrow \text{IU} & & \downarrow \text{IU} \\ g^{-1}(L) = (h \circ g)^{-1}(U) & & L & = & h^{-1}(U) \end{array}$$

Logic in General

$\mathcal{R} = r_1, r_2, \dots$

signature of relation symbols

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$\forall x, \exists x$

quantification
over individuals

$\varphi \wedge \psi, \varphi \vee \psi, \neg \psi$

boolean
operations

$r(x_1, \dots, x_n)$

relations from
the signature

$x=y$

equality

First Order Logic (FO)

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First Order Logic (FO)

$\forall X, \exists X$

quantification
over sets

$x \in X$

membership

Monadic Second Order Logic (MSO)

Logic in General

$$\mathcal{R} = r_1, r_2, \dots$$

signature of relation symbols

$$S = (\mathcal{U}, r^S_1, r^S_2, \dots)$$

relational structure

universe

interpretation of
relation symbols

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relational structure

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relation symbols

$S, u_1, \dots, u_n, \mathcal{U}_1, \dots, \mathcal{U}_m \models \varphi(x_1, \dots, x_n, X_1, \dots, X_m)$ the formula φ is true in S when

- x_i is mapped to the element u_i
- X_j is mapped to the set $\mathcal{U}_j$

Logic for Words (the precedence model)

$$\Sigma = \{a, b, c, \dots\}$$

finite alphabet

$$x \leq y$$

position x is
before position y

$$a(x)$$

position x has
letter a

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$$\mathcal{L}(\varphi) = \{ w \in \Sigma \mid w \models \varphi \}$$

language of a sentence

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language of a sentence

$\mathcal{L} \subseteq \Sigma^*$ is FO (MSO) definable iff $\mathcal{L} = \mathcal{L}(\varphi)$ for φ a FO (MSO) sentence

Some FO-definable Languages

$$x+1=y \stackrel{\text{def}}{=} \underbrace{x \leq y \wedge \neg x=y}_{x < y} \wedge \forall z . x \leq z \wedge z \leq y \Rightarrow x=z \vee z=y$$

Some FO-definable Languages

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$$a^*bc^* \quad \exists x . b(x) \wedge \forall y . y < x \Rightarrow a(x) \wedge \forall y . x < y \Rightarrow c(x)$$

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$$\text{first}(x) \stackrel{\text{def}}{=} \forall y . x \leq y \quad \text{last}(x) \stackrel{\text{def}}{=} \forall y . y \leq x$$

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$$\text{first}(x) \stackrel{\text{def}}{=} \forall y . x \leq y \quad \text{last}(x) \stackrel{\text{def}}{=} \forall y . y \leq x$$

$$(ab)^+ \quad \exists x . \text{first}(x) \wedge a(x) \wedge \forall y . a(y) \Rightarrow \neg \text{last}(y) \wedge b(y+1) \wedge \forall y . b(y) \wedge \neg \text{last}(y) \Rightarrow a(y+1)$$

Some MSO-definable Languages

$$X \subseteq Y \stackrel{\text{def}}{=} \forall x . x \in X \Rightarrow x \in Y$$

$$X \subseteq a \stackrel{\text{def}}{=} \forall x . x \in X \Rightarrow a(x)$$

$$\text{sing}(X) \stackrel{\text{def}}{=} X \neq \emptyset \wedge \forall Y . Y \subseteq X \Rightarrow Y = \emptyset \vee Y = X$$

$$X \leq Y \stackrel{\text{def}}{=} \forall x \forall y . x \in X \wedge y \in Y \Rightarrow x \leq y$$

Some MSO-definable Languages

$(aa)^+$ all words of non-zero even length over the alphabet $\Sigma = \{a\}$

$$\exists X . \forall x . first(x) \Rightarrow x \in X \wedge last(x) \Rightarrow x \notin X \wedge \forall y . x=y+1 \Rightarrow (x \in X \Leftrightarrow y \notin X)$$

all words having a on even positions over the alphabet $\Sigma = \{a, b\}$

$$\exists X . \forall x . first(x) \Rightarrow x \in X \wedge \forall y \forall z . y \in X \wedge z=y+1 \Rightarrow z \notin X \wedge X \subseteq a$$

As we shall see later, these languages are not FO-definable ...

Logic for Words (the successor model)

$$\Sigma = \{a, b, c, \dots\}$$

finite alphabet

$$x+1=y$$

position y is
the successor of x

$$a(x)$$

position x has
letter a

Makes a difference for FO-definability but not for MSO-definability

$$x \leq y \stackrel{\text{def}}{=} \forall X . (x \in X \wedge (\underbrace{\forall y \forall z . y \in X \wedge y+1=z \Rightarrow z \in X}_{\text{closed under successors}})) \Rightarrow y \in X$$

closed under successors

MSO-definability = recognisability for words

Theorem [Trakhtenbrot-Büchi-Elgot]

Let Σ be a finite alphabet and $L \subseteq \Sigma^*$ be a language.

L is recognisable $\Leftrightarrow L$ is MSO-definable.

" $\Rightarrow$ " L is recognisable $\Rightarrow L$ is the language of a finite automaton A with n states

$\exists X_1 \dots \exists X_n$. the sets $X_1 \dots X_n$ describe an accepting run of A

$$\forall x . \bigvee_{1 \leq i \leq n} x \in X_i \wedge \bigwedge_{1 \leq i < j \leq n} \neg \exists x . x \in X_i \wedge x \in X_j$$

$$\forall x . \text{first}(x) \Rightarrow \bigvee_{i \in I} x \in X_i \qquad \forall x . \text{last}(x) \Rightarrow \bigvee_{i \in F} x \in X_i$$

$$\forall x . \bigwedge_{a \in \Sigma} \bigwedge_{1 \leq i \leq n} x \in X_i \wedge a(x) \wedge \neg \text{last}(x) \Rightarrow \bigvee_{q_i \xrightarrow{a} q_j} x+1 \in X_j$$

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each MSO formula can be written as $\varphi(x_1 \dots x_n)$

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each MSO formula can be written as $\varphi(X_1 \dots X_n)$

the language of $\varphi(X_1 \dots X_n)$ is the set of words w

over the alphabet $\Sigma \times \{0,1\}^n : w, \mathcal{U}_1, \dots, \mathcal{U}_n \models \varphi(X_1 \dots X_n)$

where $i \in \mathcal{U}_j \Leftrightarrow$ the j -th bit on position i is 1

X_1	0	1	...	0
X_2	1	0	...	1
$\vdots$	$\vdots$	$\vdots$		$\vdots$
X_n	0	0	...	1
Σ	a_1	a_2	...	a_m

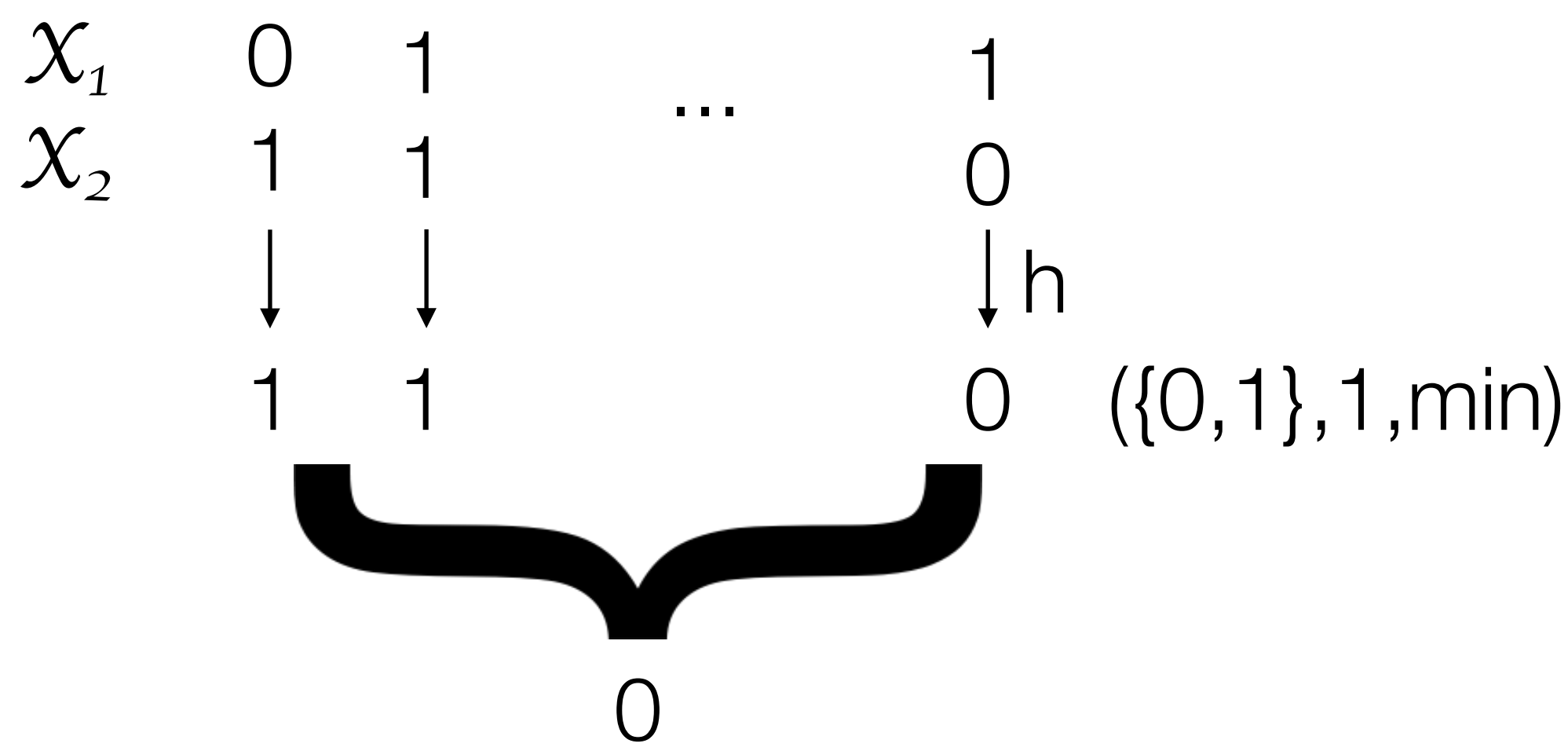
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$\mathcal{X}_1 \subseteq a$

similar construction

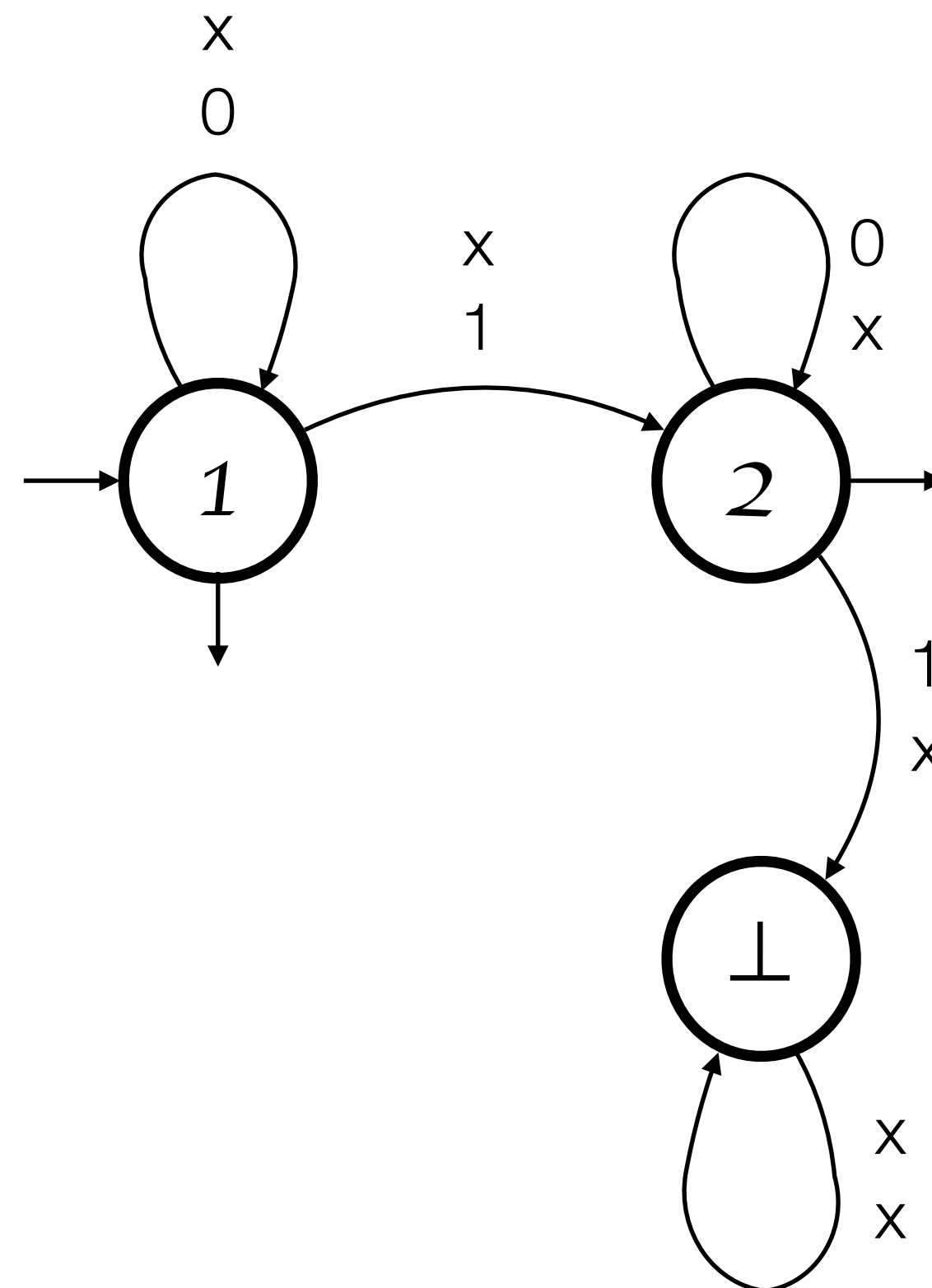
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" $\Leftarrow$ " $\varphi \wedge \psi, \varphi \vee \psi, \neg \psi$ closure of recognizable sets under boolean operations

$$\begin{array}{ccc}
 (\Sigma \times \{0,1\}^{n-1})^* & \xleftarrow{\pi} & (\Sigma \times \{0,1\}^n)^* \xrightarrow{h} M \\
 \text{IU} & & \text{IU}
 \end{array}$$

$$\varphi(x_1 \dots x_n) = h^{-1}(N)$$

$$\begin{array}{ccc}
 (\Sigma \times \{0,1\}^{n-1})^* & \xrightarrow{H(w)=\{h(v) \mid \pi(v)=w\}} & 2^M \\
 \text{IU} & & \text{IU}
 \end{array}$$

$$\underbrace{\exists x_n . \varphi(x_1 \dots x_n)}_{\pi(h^{-1}(N))} = H^{-1}(\{P \subseteq M \mid P \cap N \neq \emptyset\})$$